INTERNSHIP TASK 1 • APP CONCEPT & SCREEN FLOW

BUSBRIDGE

Your Bus. Your Route. Your Time.

Public Bus Information Gap Solver

A Mobile Application Concept for Chennai Public Transportation



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Concept and prototype proposal only. BusBridge is not an officially integrated Chennai MTC application.

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> **Document status**
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> This document is a concept and UI/UX prototype proposal prepared for the SWYNEX Technologies App Development Internship (Task 1). It does not describe a deployed system, an official partnership, or verified transport data. All route names, stop names, times, report counts and crowd levels used in examples are illustrative placeholders. Screenshots of the Figma prototype screens are included in the Appendix.

1. Executive Summary

BusBridge is a proposed mobile application concept designed to improve the accessibility and usability of public bus information in **Chennai**. It aims to bring route discovery, stop information, estimated arrival details, community-reported updates and journey assistance into one simple, passenger-friendly application.

The concept focuses on helping passengers to:

- Find bus routes between two locations.
- Discover nearby bus stops.
- Understand available bus services.
- View estimated arrival information when available.
- Check community-reported crowd conditions.
- Report delays, cancellations and route-related issues.
- Save frequently used routes and receive relevant alerts.

> **Concept status**
>
> BusBridge is currently a concept and prototype proposal. It is not an officially integrated Chennai MTC application, and it does not claim live GPS tracking or access to official transport data. Real-time features would depend on authorized access to reliable data feeds.

2. Introduction

Public buses are one of the most affordable and widely used modes of transport in Chennai. College students, office employees, daily commuters, senior citizens and occasional travellers all rely on bus services to reach classes, workplaces, markets and homes every day.

For a regular passenger on a familiar route, the journey is simple. For a new passenger, or for someone travelling to an unfamiliar part of the city, the experience can be less clear. Knowing which bus to take, where to board, which stops lie ahead and when the next bus may arrive often requires asking other people or checking several different sources.

Accessible and understandable bus information can make these journeys easier and more predictable. **BusBridge** is proposed as a solution focused on reducing these information gaps during everyday bus journeys, with simple language, clear screens and honest labelling of where each piece of information comes from.

3. Problem Statement

The following are passenger information challenges that BusBridge aims to address. They are presented as common difficulties, not as a claim that every Chennai bus service lacks digital information.

#	Passenger Challenge	What it means for the passenger
1	Identifying the correct route	Passengers may find it difficult to work out which bus to board for a given destination.

#	Passenger Challenge	What it means for the passenger
2	Understanding route connections	New passengers may struggle with route connections and intermediate stops.
3	Reliable arrival information	Finding dependable arrival-time information can be difficult.
4	Delays and service changes	Passengers may not learn about unexpected delays or changes until they are already waiting.
5	Crowd conditions	Information about how crowded a bus is may not be available before boarding.
6	Multiple information sources	A single journey may require checking several different sources.
7	Unfamiliar areas	People new to an area may find it hard to identify suitable stops and routes.

> **Problem Statement**
>
> "There is a need for a simple and accessible platform that brings public bus route information, estimated arrival details, passenger-reported updates, and journey assistance together in one place."

4. Proposed Solution

BusBridge addresses the challenges above through a focused set of passenger-facing features:

Bus Route Search

Find routes from a source to a destination.

Source and Destination

Simple From / To selection with a swap option.

Nearby Bus Stop Discovery

Find stops close to the passenger, or search manually.

Route and Stop Information

View stops along a route and choose boarding and exit stops.

Estimated Arrival Information

Shown only where supported, with a last-updated time.

Community Crowd Reporting

Low, Medium or High reports submitted by passengers.

Passenger Issue Reporting

Report delays, cancellations and route problems.

Saved Routes and Notifications

Save favourite routes and receive relevant service alerts.

4.1 Three Types of Information

To keep the application honest and easy to trust, BusBridge separates information into three clearly labelled categories. Every route, arrival and crowd item shown in the app carries one of these labels.

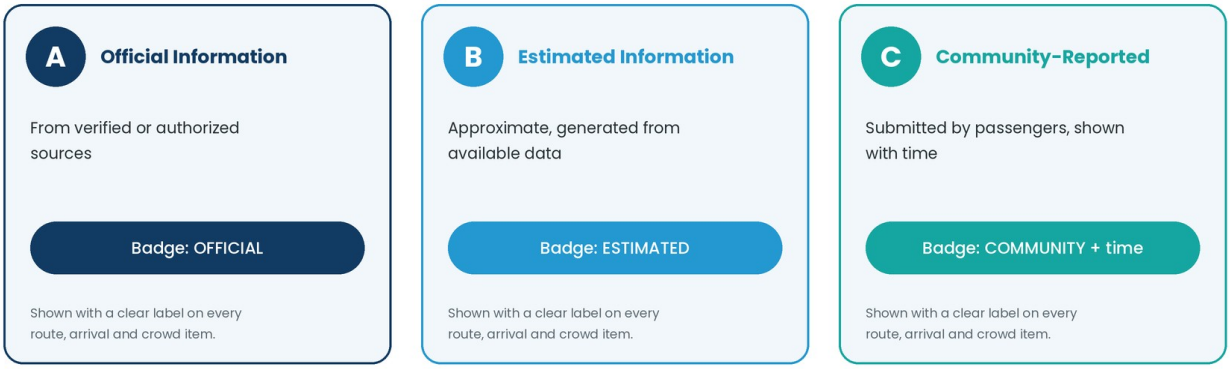


Figure 1: BusBridge information-source labels

> **Real-time tracking depends on data access**
>
> Live bus tracking requires access to reliable bus GPS feeds or an authorized data source. BusBridge does not pretend that live GPS tracking or official MTC API integration already exists. Where such data is unavailable, the app shows a clear “unavailable” status rather than a fabricated position.

5. Project Objectives

1. **Simplify bus route discovery** for passengers travelling between two locations.
2. **Improve access to public transportation information** through one easy-to-use application.
3. **Help passengers plan journeys more efficiently** with route, stop and timing information.
4. **Make crowd-related information easier to understand** using three simple levels.
5. **Allow passengers to report service-related issues** such as delays and cancellations.
6. **Improve the usability of bus information for new commuters** through clear language and layout.
7. **Build a scalable concept** that can potentially expand to other cities in the future.

6. Scope of the First Version (MVP)

The first version of BusBridge is intentionally small and focused. It is a **concept and UI/UX prototype** for Task 1, with a possible working MVP only in later phases. Nothing in this document has been developed, deployed or connected to any live data source.

Scope Area	First Version (MVP) Decision
Geography	Chennai only. Route, stop and landmark information would initially be limited to Chennai, Tamil Nadu. Other cities are future possibilities.
Platform	Android-first mobile application concept; adaptable to iOS later.

Scope Area	First Version (MVP) Decision
Screens	Six proposed screens: Home Dashboard, Search Bus and Route, Bus Route Details and Arrival Information, Crowd Status, Report Bus Issue, and Saved Routes and Alerts.
Languages	English and Tamil proposed as language options (see “Language Support: Tamil and English”).
Data	Verified public route information and manually maintained prototype data. Official arrival data and live GPS are included only if authorized access becomes available.
Community features	Passenger crowd reports and issue reports, always labelled as community-submitted.
Out of scope for the MVP	Ticketing or payments, official MTC integration, guaranteed live tracking, multi-city coverage, and any claim of verified schedules.

The six screens in the MVP map to the features and requirements described in the following sections, and each can be designed and reviewed on its own in Figma.

7. Target Users

BusBridge is designed for people who depend on public buses in different ways. Each group has distinct needs, which shaped the screens and features in this proposal.

Target User	Needs	How BusBridge Supports Them
College Students	Affordable and convenient bus journey planning	Students travel on a budget and often on fixed timetables. Route search and saved routes help them plan daily trips to and from campus.
Office Employees	Finding routes and planning daily commutes	Employees need repeatable, dependable commutes. Favourite routes and alerts reduce repeated searching.
Daily Commuters	Quick access to frequently used routes	Recent searches and saved routes on the Home Dashboard give one-tap access to regular journeys.
New Passengers	Understanding bus routes and stops	Stop-by-stop route views and clear boarding / destination markers explain how a route works.
Senior Citizens	Simple and readable travel information	Large text, high contrast and uncluttered screens make information easy to read and use.
Occasional Travelers	Discovering suitable routes in unfamiliar areas	Nearby stops and manual location search help visitors find a starting point and a suitable route.

8. Key Features

8.1 Bus Route Search

- Enter a source and a destination.
- View the available route options.
- Display bus numbers and intermediate stops where data is available.
- Show estimated journey duration when supported.

8.2 Nearby Bus Stops

- Discover bus stops near the passenger.
- Display stop names.
- Allow passengers to select a stop to see related route information.
- Location permission is optional; **manual location search** is always available as a fallback.

8.3 Bus Arrival Information

- Display estimated arrival information only when reliable data is available.
- Clearly show the **last updated time**.
- Show whether the information is **Official, Estimated or Unavailable**.
- Never display fabricated live bus positions.

8.4 Crowd Status

Passengers see a simple three-level indicator, and can submit their own report:

LOW	MEDIUM	HIGH
Seats are available.	Mostly full; some standing.	Very crowded; limited space.

Each report shows its **timestamp**, the **number of recent reports** and a **freshness indicator**, and is labelled **community-submitted**. A note reminds passengers that reports may become outdated.

8.5 Report Bus Issues

Passengers can report: **bus delay**, **bus cancellation**, **route change**, **incorrect route information**, and **other** service-related issues. Sharing location is **optional**, and no personal details are required.

8.6 Saved Routes and Alerts

- Save frequently used routes.
- View recent journeys.
- Receive relevant service alerts when supported.
- Manage favourite routes, including removing routes that are no longer needed.

9. Functional Requirements

The table lists the functional requirements for the MVP concept. Priority reflects the intended order of design and development.

ID	Requirement	Description	Priority	Screen
FR-01	Bus route search	The passenger can enter a source and destination, swap them, and view available route options with bus number, route name and estimated duration where supported.	High	2
FR-02	Nearby stop discovery	The app can list nearby stops when location permission is granted, and offers manual location search when it is not.	High	1
FR-03	Route and stop details	The passenger can view the stops on a selected route and mark boarding and destination stops.	High	3
FR-04	Arrival information with source label	Arrival estimates are shown only where reliable data exists, with a source label (Official, Estimated or Unavailable) and a last-updated time. Live GPS location is shown only if authorized data is available.	High	3
FR-05	Community crowd reporting	Passengers can submit a Low, Medium or High crowd report and view recent reports with timestamps, report counts and freshness indicators, all labelled community-submitted.	Medium	4
FR-06	Issue reporting	Passengers can report a delay, cancellation, route change, incorrect information or other issue, with optional location sharing, validation and a confirmation message.	Medium	5
FR-07	Saved routes, alerts and settings	Passengers can save and remove routes and stops, view recent journeys and service alerts, and manage notification and language settings.	Medium	6

10. Non-Functional Requirements

These requirements describe the quality of the experience. Values marked as targets are design goals to be validated during prototyping, not measured results.

Category	Requirement
Usability	Common tasks (search a route, save a route, submit a report) should take only a few taps. Labels use plain language, and every icon is paired with text. The app should be usable by first-time passengers without instructions.
Performance	Screens should respond quickly on typical mobile connections. Target: route results appear within a few seconds, with a loading indicator shown while waiting. Recently viewed and saved routes load from local storage where possible.

Category	Requirement
Accessibility	Text contrast of at least 4.5:1, minimum body text of 14 sp, support for larger system font sizes, and touch targets of at least 48 dp (above the 24 × 24 CSS pixel minimum in WCAG 2.2). Status is never shown by colour alone.
Privacy	Location is optional and requested only when needed. Reports do not require a name, phone number or account. Only the minimum data needed is collected, and the app explains how data is used.
Reliability	Every piece of information carries a source label and last-updated time. Missing data is shown as unavailable, never guessed. Failed actions show clear messages and can be retried.
Responsiveness	Layouts adapt to common Android screen sizes and font scales (base frame 360 × 800 dp), and work in portrait orientation first. Tamil text and longer labels must not break layouts.

11. App Screen Descriptions (Six Screens)

The six screens below form the core prototype. Each description is written so that a UI/UX designer can build the screen in Figma. All names, times and statuses in the "Example content" rows are placeholders only.

11.1 Shared Design Specification

Item	Specification
Frame	Android mobile frame, 360 × 800 dp (adaptable to iOS)
Grid and spacing	8 dp spacing system; 16 dp side margins; 12–16 dp card corner radius
Colours	Navy #123B63 (headers, primary text on light), Sky #2498D1 (links, selected state), Teal #16A6A1 (primary buttons, positive states), Light #F2F7FB (card backgrounds), Charcoal #263238 (body text)
Typography	Poppins Bold / SemiBold for titles; Inter or Arial for body. Minimum body size 14 sp; buttons 16 sp
Bottom navigation	Four tabs on main screens: Home, Routes, Report, Saved. Active tab uses Teal with a label
Source labels	Small pills: OFFICIAL (Navy), ESTIMATED (Sky), COMMUNITY (Teal) with a "Updated hh:mm" timestamp
Accessibility	Contrast ≥ 4.5:1, touch targets ≥ 48 dp, icons always paired with text labels
Languages	English and Tamil (தமிழ்) text must fit every component; allow about 30–40% extra text length and avoid fixed-width labels

SCREEN 1 | Home Dashboard	
Purpose	Provide quick access to the main BusBridge features and act as the starting point of every journey.
Main UI components	• BusBridge logo and application name in the top bar. • Greeting message, for example "Good morning! Where are you travelling today?" • Search destination field, with "From" and "To" location inputs. • Quick actions: Search Bus, Nearby Stops, Saved Routes. • Nearby bus stops section (horizontal cards). • Favourite routes section. • Recent searches list. • Bottom navigation bar.
Layout description	Navy header band (about 160 dp) with rounded bottom corners holds the logo and greeting. A white search card overlaps the header and contains the From and To fields and the Search Bus button. Below, content scrolls vertically: quick-action chips, Nearby Stops cards, Favourite Routes, Recent Searches. The bottom navigation bar stays fixed.

Buttons and interactions	• **Search Bus** (Teal, full width): opens Screen 2 with the entered From / To. • **Nearby Stops**: asks for optional location permission; if declined, opens manual location search. • **Saved Routes**: opens Screen 6. • Tapping a favourite or recent search pre-fills Screen 2.
Navigation destination	Search Bus → Screen 2. Saved Routes → Screen 6. Bottom navigation tabs → corresponding screens.
Example content	From: Chennai Central | To: T. Nagar. Nearby: “Sample Stop A – approx. 150 m”. Favourite: “Home to College (sample)”. Recent: “Chennai Central → T. Nagar”.
Design considerations	Clear spacing, simple icons paired with text, readable type, and a visually dominant search section. Empty states (no favourites yet) show a short helpful message.

SCREEN 2 | Search Bus and Route

Purpose	Help passengers find suitable bus routes between two locations.
Main UI components	• Source location field and destination location field. • Swap-locations button between the two fields. • Search button. • List of available bus route cards, each with bus number, route name, intermediate-stop summary and estimated journey duration (when supported).
Layout description	Compact navy app bar with a back arrow and the title “Search Bus”. Directly below, a card holds the From and To fields stacked vertically with the swap icon aligned on the right. A Search button follows. The results list appears beneath as vertically stacked cards with 12 dp spacing.
Buttons and interactions	• **Swap** exchanges source and destination. • **Search** loads route options, with a loading state and an “No routes found” empty state. • Typing in a field shows location suggestions; a “Use current location” option appears only if permission is granted. • Tapping a route card selects it.
Navigation destination	Selecting a route card → Screen 3 (Bus Route Details). Back arrow → Screen 1.
Example content	Card: “Bus [No.] – Sample Route A”, “Via: Stop 1, Stop 2, Stop 3”, “Approx. 45 min (estimated)”, pill: ESTIMATED. (Bus numbers are placeholders.)
Design considerations	Route cards show only the essentials so they scan quickly. The information-source pill is always visible. Duration is hidden, not guessed, when not supported.

SCREEN 3 | Bus Route Details and Arrival Information

Purpose	Show the selected bus route, its stops and any available arrival information in one place. Live GPS location is a **conditional future capability**: it appears only if reliable, authorized GPS data exists.
Main UI components	• Bus number and route name. • Route map or simplified route diagram. • List of stops with selected boarding and destination stops highlighted. • Arrival estimate card, if available. • Next-stop information. • Last-updated timestamp and information-source label (Official, Estimated or Unavailable). • Live location status card (Available or Unavailable). • Action buttons: Crowd Status, Report Issue, Save Route.
Layout description	Header with the bus number badge (Navy) and route name. Below it, a simplified vertical route line (circles connected by a line) or a map preview of about 200 dp height. A stop list follows, with the boarding stop marked in Teal and the destination in Navy. An arrival card shows the estimate, next stop, source pill and “Last updated” time. A separate live-location status card sits beneath it. Three action buttons close the content.
Buttons and interactions	• **Crowd Status** → Screen 4. • **Report Issue** → Screen 5. • **Save Route** (bookmark icon): toggles saved state and shows a confirmation toast. • Tapping a stop sets it as the boarding or destination stop. • **Refresh** (pull down) reloads arrival data.
Navigation destination	Crowd Status → Screen 4. Report Issue → Screen 5. Save Route → adds to Screen 6. Back → Screen 2.
Example content	“Boarding: Chennai Central”, “Destination: T. Nagar”, “Next stop: Sample Stop 2”, “Arrival: Estimated 6 min – Updated 10:42 (ESTIMATED)”, “Live location: currently unavailable”. All values are samples.
Unavailable state	**When reliable GPS data is not available**, show a neutral grey card with the message **“Live location currently unavailable”**, the time it was last checked, and a short explanation such as “Live bus positions need an authorized data source.” No bus marker is drawn on the map. If arrival data is also missing, show “Arrival information unavailable” with the scheduled route and stops only.
Design considerations	Never show a fake or estimated bus position as if it were live. If authorized GPS data becomes available in future, the live-location card turns into a map marker with an OFFICIAL pill and an updated time. The stop list must stay readable on small screens; long English and Tamil stop names wrap rather than truncate.

SCREEN 4 | Crowd Status

Purpose	Help passengers understand recently reported crowd conditions before boarding, and contribute their own report.
Main UI components	• Selected bus number or route. • Crowd status indicator showing the most recent level. • **Recent report count**, for example “5 reports in the last 30 min”. • **Report freshness indicator** for the latest report (Fresh, Ageing or Old). • Low / Medium / High selection options. • “Report crowd status” button. • Recent passenger reports list, each with a **timestamp** and a “Community-submitted” label. • Last-reported time.
Layout description	Top section shows the bus badge and route name. A large indicator card (about 140 dp) shows the latest level in text and colour (Teal = Low, Amber = Medium, Red = High), the last-reported time, the recent report count and a freshness pill. Below are three selectable option chips, the Report button, and a list of recent reports. The community note sits at the bottom of the screen.
Freshness indicators	Proposed thresholds, to be tuned during testing: • **Fresh:** reported within the last 15 minutes (Teal pill). • **Ageing:** 15 to 60 minutes ago (Amber pill). • **Old:** more than 60 minutes ago (Grey pill, faded text).
Buttons and interactions	• Tap a chip to choose Low, Medium or High. • **Report crowd status** submits the choice and shows “Thanks, your report was added”. • Pull to refresh the report list. • Repeated or too-frequent submissions from the same device are limited.
Navigation destination	Back → Screen 3. Bottom navigation available.
Example content	“Latest: MEDIUM – reported 8 min ago · FRESH · COMMUNITY-SUBMITTED”. “5 reports in the last 30 min”. List: “HIGH – 25 min ago (Ageing)”, “LOW – 1 hr 20 min ago (Old, may be outdated)”. All values are samples.
No-report state	If there are no recent reports, show “No crowd reports yet for this bus” and invite the passenger to submit the first one.
Design considerations	Every report, count and level is labelled **“Community-submitted”**. Always display this note: **“Crowd levels are community-reported and may not reflect current conditions.”** Never rely on colour alone: each level and freshness state has a text label and an icon.

SCREEN 5 | Report Bus Issue

Purpose	Allow passengers to report problems related to bus services quickly, with clear privacy information and confirmation.
Main UI components	• Issue type selector: Delay, Cancellation, Route change, Incorrect information, Other. • Bus number input. • Route selection. • Description field (multi-line, 250-character limit with a live counter). • Optional image attachment. • **Optional location sharing toggle** (off by default). • **Privacy information** note with a “How your report is used” link. • Submit report button. • **Submission confirmation** state.
Layout description	Single-column form on a light background. Issue types appear as selectable chips at the top, followed by Bus number, Route dropdown, Description, the optional attachment tile and the location toggle. The privacy note sits just above a Submit button fixed to the bottom of the screen. After submission, the form is replaced by a confirmation card with a check icon.
Form validation	• Issue type is **required**; Submit stays disabled until one is chosen. • At least one of Bus number or Route must be provided. • Description is optional but limited to 250 characters; the counter turns amber near the limit. • Attachments must be an image and within a size limit; unsupported files show an inline message. • Error messages appear under the relevant field in plain language.
Optional location sharing	• The toggle is **off by default**. Turning it on asks for location permission only if it has not been granted. • Helper text: “Share your approximate location to help verify this report. You can submit without it.” • If permission is denied, the form still works and shows no error.
Privacy information	• No name, phone number or account is required. • Images are optional and must not show identifiable people. • Reports are labelled community-submitted and may be reviewed or moderated. • A short note explains what is collected and why.
Submission confirmation	A confirmation card shows “Report submitted”, a short summary of the issue type and route, a sample reference such as “Report ID: [sample]”, and the buttons **Back to Home** and **Report another issue**. If submission fails, the form is kept and a retry message is shown (see Error and Empty States).
Navigation destination	Submit → Confirmation state → Screen 1 (Home). Back → previous screen.
Example content	Issue: Delay. Bus: “[Bus No.]”. Route: “Sample Route A”. Description: “Bus has not arrived for a long time at this stop.” Location sharing: Off.
Design considerations	Keep the form short and low effort. Use clear labels in English and Tamil, large touch targets and visible focus states. Do not collect more data than needed.

SCREEN 6 | Saved Routes and Alerts	
Purpose	Allow passengers to manage their frequently used routes, stops and alerts in one place.
Main UI components	• Favourite routes. • Saved bus stops. • Recent journeys. • Service alerts. • Notification settings. • Language selection (English / Tamil). • Remove saved route option.
Layout description	Simple list-based layout with a segmented control at the top: Routes | Stops | Alerts. Each list row has a route name, source / destination text and a trailing overflow menu. Recent journeys appear beneath the favourites. Notification and language settings are reached from a gear icon in the top bar.
Buttons and interactions	• Tap a row to open Screen 3 for that route. • **Remove** (swipe left or overflow menu) with an undo snackbar. • Notification toggles: service alerts on / off, alert types. • **Language** selector: English or Tamil, applied immediately. • Tap an alert to see its details and source label.
Navigation destination	Saved route row → Screen 3. Back / bottom navigation → Home or other tabs.
Example content	Favourite: “Home to College (sample)”. Stop: “Sample Stop A”. Alert: “Sample alert: possible delay on a saved route – ESTIMATED, updated 09:15”.
Design considerations	Clear route names, comfortable row height (at least 64 dp), and easy-to-use controls. Empty states explain how to save a first route. Alerts appear only when supported by available data.

12. Error and Empty States

Clear messages keep passengers informed when something is missing or goes wrong. Each state below should be designed as a simple card with an icon, a short message and one clear action. The wording is a proposal and should be available in English and Tamil.

State	When it appears	What the passenger sees	Recovery action
No routes found	Search returns no routes between the chosen places.	“No routes found for these locations.”	Check spelling, try a nearby stop or landmark, or swap source and destination.
Network failure	The device is offline or the server cannot be reached.	“No internet connection.” Saved routes and recent searches remain visible where available.	Retry button; data refreshes when the connection returns.

State	When it appears	What the passenger sees	Recovery action
Location permission denied	The passenger declines location access.	“Location is off. You can still search by typing a place.”	Manual location search, with an option to open settings later.
Unavailable live data	No authorized live or arrival data exists for a route.	“Live location currently unavailable” or “Arrival information unavailable”, with the last-checked time.	Show route and stops only; offer to save the route and check again later.
No crowd reports	No recent reports exist for the selected bus.	“No crowd reports yet for this bus.”	Invite the passenger to submit the first report.
Failed report submission	A report cannot be sent because of a network or server problem.	“We couldn’t send your report.” The form and the entered text are kept.	Retry button; the report is not lost, and a Save draft option can be considered.

13. App Navigation Flow

Passengers move from the Home Dashboard to a route search, select a route, and then reach the Bus Route Details screen. From there, they can view arrival information (and live location only where authorized data exists), check crowd status, report an issue, or save the route. Saved Routes and Alerts can also be reached directly from the Home Dashboard or the bottom navigation bar.

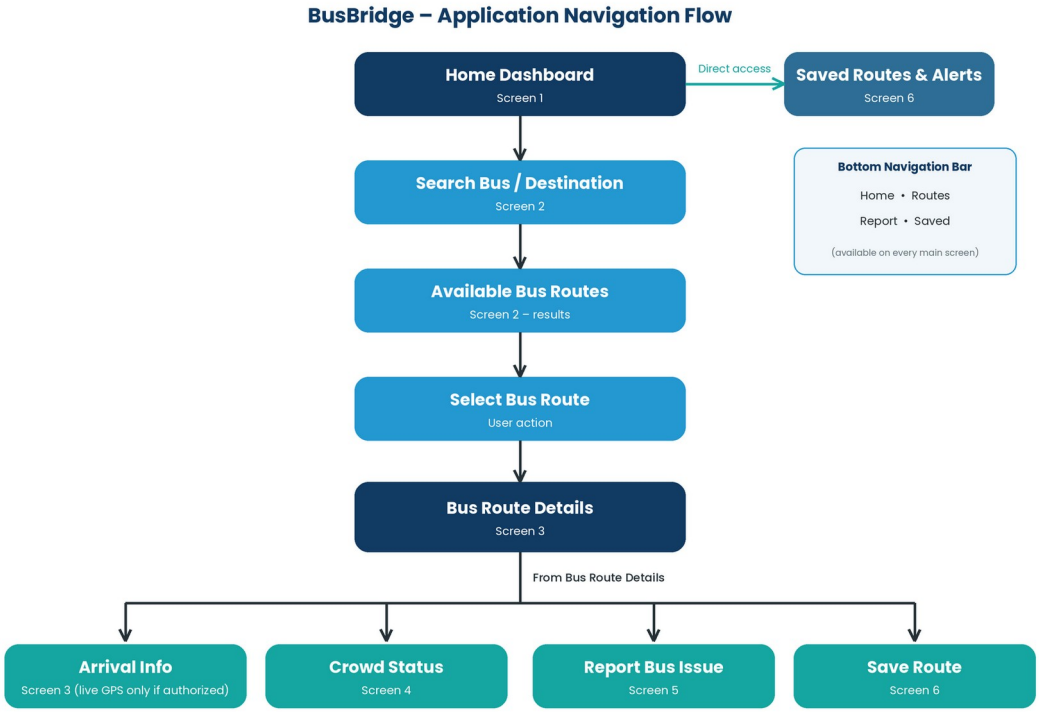


Figure 2: BusBridge navigation flow

From	Action	To
Home Dashboard	Search Bus	Search Bus / Destination (Screen 2)
Search Bus	Select a route card	Bus Route Details (Screen 3)
Bus Route Details	Crowd Status	Crowd Status (Screen 4)
Bus Route Details	Report Issue	Report Bus Issue (Screen 5)
Bus Route Details	Save Route	Saved Routes and Alerts (Screen 6)
Home / Bottom navigation	Saved	Saved Routes and Alerts (Screen 6)

14. User Journey Example

Scenario: A college student wants to travel from **Chennai Central** to **T. Nagar**.

Step	Student Action	BusBridge Response
1	Opens BusBridge.	Home Dashboard appears with search card, nearby stops and recent searches.
2	Enters Chennai Central as the source.	Location suggestions appear; the student selects the source.
3	Enters T. Nagar as the destination.	Destination is set; Search Bus becomes active.
4	Taps Search Bus.	Route options are shown based on available route data, each with a source label.
5	Selects a suitable route.	Bus Route Details opens with the stop list and highlighted boarding and destination stops.
6	Checks route stops and available arrival information.	Shows stops, next stop and arrival estimate with its label and last-updated time, or “unavailable”.
7	Checks the latest crowd report.	Crowd Status shows the most recent community report with its timestamp and the “may be outdated” note.
8	Saves the route.	Route is added to Saved Routes; a confirmation message is shown.

> **Sample journey only**
>
> This is an illustrative journey. It does not use actual MTC bus numbers, live arrival times or verified schedules.

15. Language Support: Tamil and English

English and Tamil (தமிழ்) are proposed as the two language options. Tamil-language usability is an **important consideration for Chennai passengers**, because many passengers are more comfortable reading bus information in Tamil, and route boards and stop names are commonly read in Tamil.

Consideration	Proposed Approach
Language choice	Offer English and Tamil on first launch and in settings (Screen 6). Changing the language applies immediately without restarting.
Bilingual place names	Store stop and landmark names in both English and Tamil where verified data exists, and allow search in either language.
Script and fonts	Use a font with full Tamil script support and test that conjunct characters display correctly on common Android devices.

Consideration	Proposed Approach
Layout flexibility	Tamil labels can be longer than English labels. Buttons, chips and cards must wrap or grow rather than truncate.
Plain, consistent wording	Use short, everyday words for actions and statuses (for example Low, Medium, High) and keep terms consistent across screens.
Quality review	Translations should be reviewed by fluent Tamil speakers, and the prototype tested with Tamil-speaking passengers, before release.

> **Proposed, not implemented**
>
> Tamil and English are proposed language options for the concept. No translations or localisation work have been completed yet. Voice input and additional languages are listed under Future Enhancements.

16. Technology and Tools

This section proposes **one primary stack** for later implementation. It is a plan, not an existing system. Design tools used for Task 1 are listed separately from the technologies proposed for building the app. The final selection may depend on project requirements and internship guidance.

16.1 Design Tools (Task 1)

Tool	Purpose
Figma	Design the six screens, build the clickable prototype, document the navigation flow and share the design for feedback. This is the tool used for the Task 1 screens.

16.2 Proposed Implementation Stack (Later Phases)

Layer	Primary Proposed Technology	Purpose
Mobile frontend	Flutter	Build the Android-first app from one codebase, with the option to extend to iOS later.
Backend API	Node.js with Express	Provide APIs for routes, stops, reports and alerts, and apply source labels and validation.
Database	PostgreSQL	Store routes, stops, saved routes, reports and their timestamps in a structured way.
Maps	OpenStreetMap-based maps	Show stops and route diagrams without a paid map service during the prototype stage.
Notifications	Firebase Cloud Messaging	Deliver service alerts to passengers who opt in.
Authentication (optional)	Firebase Authentication	Only if saved routes need to sync across devices; reports remain possible without an account.
Version control	Git and GitHub	Track changes and support collaboration.

> **Alternatives (not selected)**
>
> React Native for the frontend, Firebase for the backend and database, and Google Maps Platform for maps are reasonable alternatives. They are noted only as fallback options if guidance or project needs change, and are not part of the proposed stack.

17. Data Sources and Reliability

BusBridge could obtain information from the following sources, subject to availability and authorization:

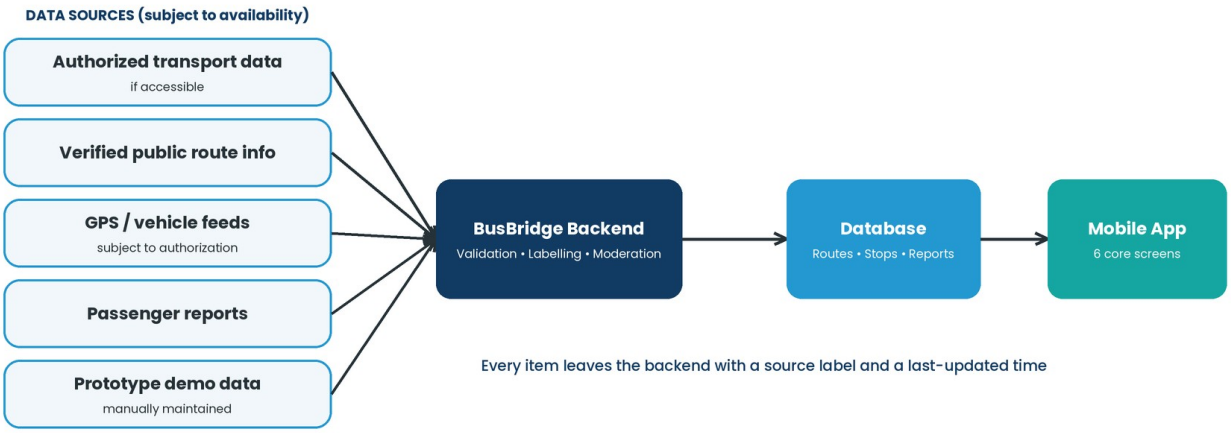


Figure 3: Proposed data flow with source labelling

Potential Source	Notes
Authorized transport data	Used only if access is granted by the relevant authority.
Verified public route information	Route and stop details from publicly available, verified material.
GPS or vehicle location feeds	Subject to authorization; not assumed to exist today.
Passenger-submitted reports	Crowd and issue reports, labelled as community-reported.
Manually maintained prototype data	Sample data used for demonstration during the internship.

17.1 Reliability Principles

1. **Never show estimated data as official data.**
2. **Show the last updated time** on every time-sensitive item.
3. **Mark passenger reports clearly** as community-reported.
4. **Show an unavailable status** when data is missing.
5. **Avoid misleading live-tracking claims.**
6. **Validate and moderate user reports** where possible.

18. Unique Value Proposition

BusBridge is useful because of how it **combines and presents** several passenger needs in one place:

- Bus route discovery.
- Passenger-friendly journey information.
- Crowd reporting.

- Bus service issue reporting.
- Saved routes and alerts.

The proposed **community-assisted information approach** lets passengers contribute useful updates, such as crowd levels and service issues, while the app clearly marks which information is official, estimated or community-reported.

> **A note on differentiation**
>
> BusBridge does not claim to be the first or only application of its kind. Its differentiation lies in the proposed combination of these features and, in particular, the clear information-source labels shown throughout the experience.

19. Future Enhancements

The following are future possibilities. None of them are completed features.

More accurate arrival estimates

Through authorized live data.

Expansion to more cities

Additional Tamil Nadu cities.

Wider language support

Tamil voice input and additional languages beyond the proposed Tamil and English options.

Accessibility improvements

Better support for all passengers.

Voice-assisted route search

Hands-free searching, including in Tamil.

Better notification customization

Finer control of alerts.

Improved report verification

More reliable community reports.

Offline access

View previously saved route information without internet.

20. Expected Benefits

Passengers

Easier route discovery.

Better journey planning.

Access to community updates.

Students

Convenient daily commuting information.

New commuters

Easier understanding of routes and stops.

Public transportation ecosystem

A potential additional channel for passenger feedback and service information.

These are potential benefits of the concept. This proposal makes no guarantee of reduced waiting times or operational improvements.

21. Limitations and Challenges

Challenge	Realistic Way to Address It
Availability of reliable transport data	Start with verified public route data and demo data; add official sources if access is granted.
Access to authorized GPS information	Seek authorization before using feeds; show “unavailable” until then.
Internet connectivity requirements	Cache recently viewed and saved routes for limited offline use; show clear offline messages.
Accuracy of community reports	Show timestamps, fade old reports, rate-limit submissions and add moderation or report-flagging.
Location permission and privacy	Keep location optional, provide manual search, collect minimal data and explain how data is used.
Data verification and maintenance	Define a review process, record data sources and update dates, and allow passengers to flag errors.
Existing official information channels	Press reports suggest MTC may already offer its own real-time information. BusBridge should complement existing services, not duplicate them, and must not assume access to them.
Keeping route information updated	Schedule regular checks, display last-verified dates and use the issue-report feature to find outdated information.

22. Conclusion

BusBridge is a practical mobile application concept intended to make Chennai public bus information easier to access and understand. By combining route discovery, stop information, estimated arrival details, crowd reporting, issue reporting and saved routes in one application, it responds directly to the passenger information challenges described in this document.

The proposed six-screen prototype focuses on **usability**, **clear information presentation** and **community participation**. Its emphasis on labelling information as official, estimated or community-reported is intended to build passenger trust and avoid misleading claims.

This concept can serve as a foundation for future implementation and expansion, beginning with a Figma prototype and growing into a working application as reliable data sources and guidance become available.

23. Research and Reference Sources

This list includes only sources that were actually consulted while preparing this revision. No user surveys, interviews or official data requests have been carried out, and no statistics from these sources are used as findings in this document.

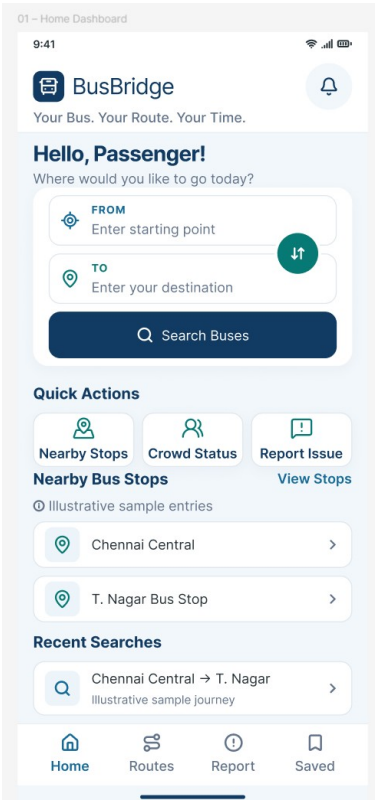
#	Source	How it was used	Accessed
1	Metropolitan Transport Corporation (Chennai), Wikipedia. https://en.wikipedia.org/wiki/Metropolitan_Transport_Corporation_(Chennai)	Background on MTC as the public bus operator in Chennai. It also lists mtcbus.tn.gov.in as MTC's website.	31 Oct 2026
2	DT Next, "Footboarding and overcrowding of buses common in Chennai". https://www.dtnext.in/amp/story/news/chennai/a-bus-load-of-problems-plagues-mtc	A search snippet indicating press coverage of crowding and of MTC's real-time tracking launch. Only the snippet was read; the full article was not reviewed, so no details are relied on.	31 Oct 2026
3	W3C, "What's New in WCAG 2.2". https://www.w3.org/WAI/standards-guidelines/wcag/new-in-22/	Reference for the accessibility guideline WCAG 2.2, including its minimum target size criterion, used for the accessibility requirements.	31 Oct 2026
4	Task brief, SWYNEX Technologies App Development Internship (Task 1)	Defines the required sections, scope and reliability rules for this document.	Provided

> **Note on sources**
>
> The official MTC website was identified as a possible source of route information but has not been used to collect or verify route data for this document. Any real route or stop data added later should be taken from verified or authorized sources and referenced here.

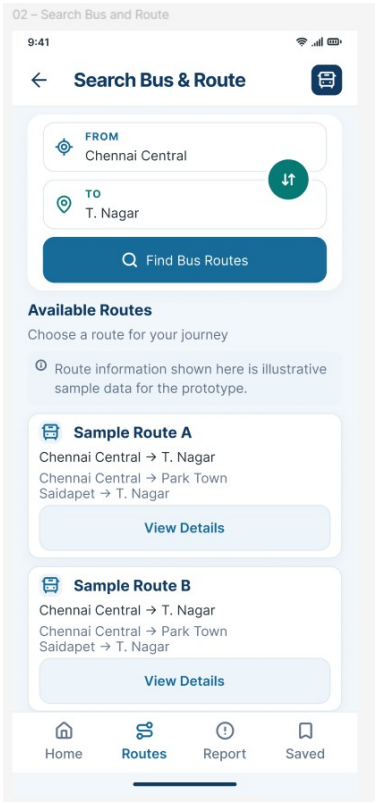
24. Appendix

A. Six-Screen Designs (Figma Screenshots)

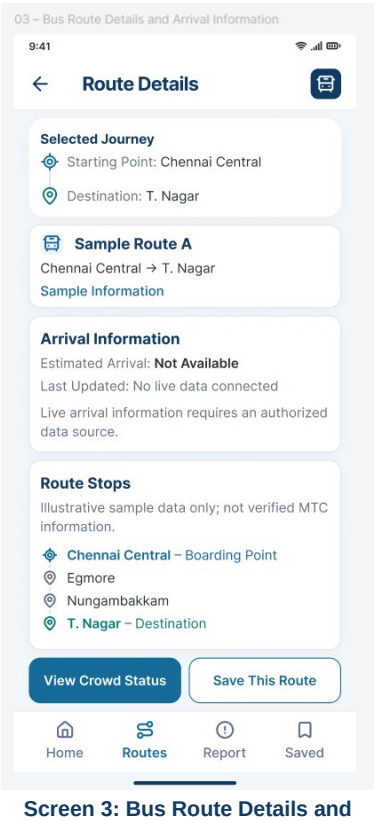
The six screens below are screenshots of the BusBridge Figma prototype. All route names, stops and reports shown are illustrative sample data for the prototype, not verified MTC information.



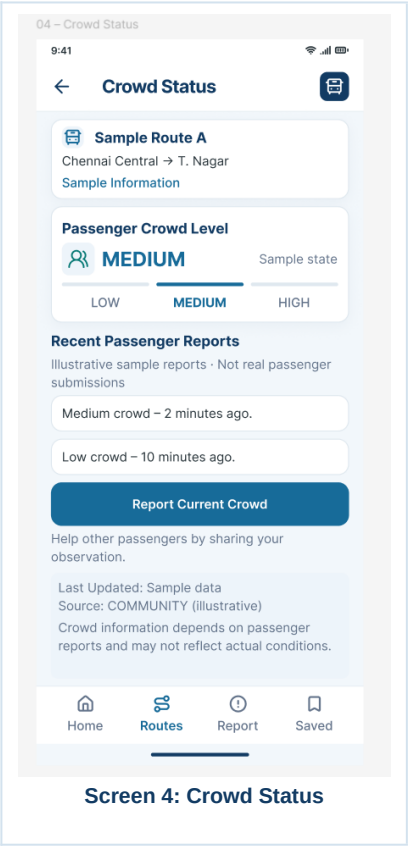
Screen 1: Home Dashboard



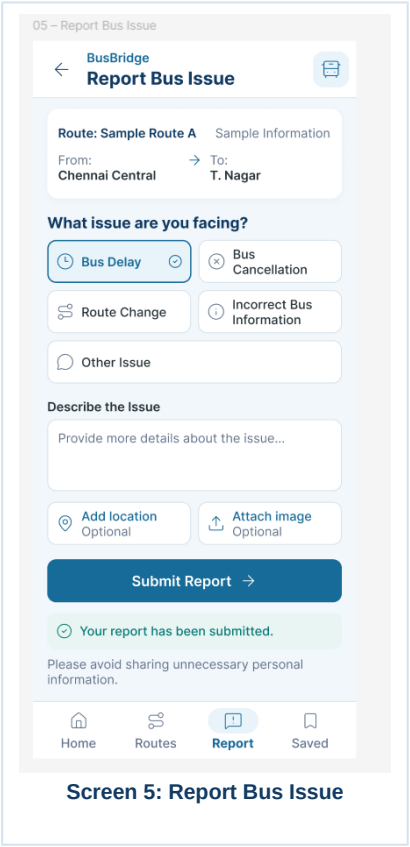
Screen 2: Search Bus and Route



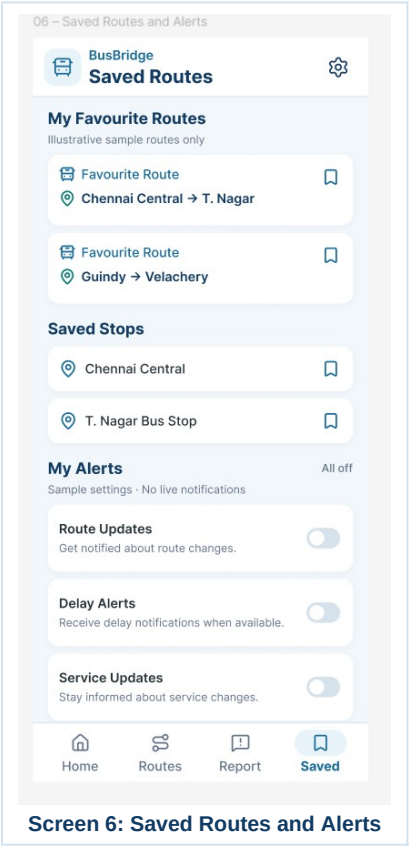
Screen 3: Bus Route Details and Arrival Information



Screen 4: Crowd Status



Screen 5: Report Bus Issue



Screen 6: Saved Routes and Alerts

B. Navigation Flow Diagram

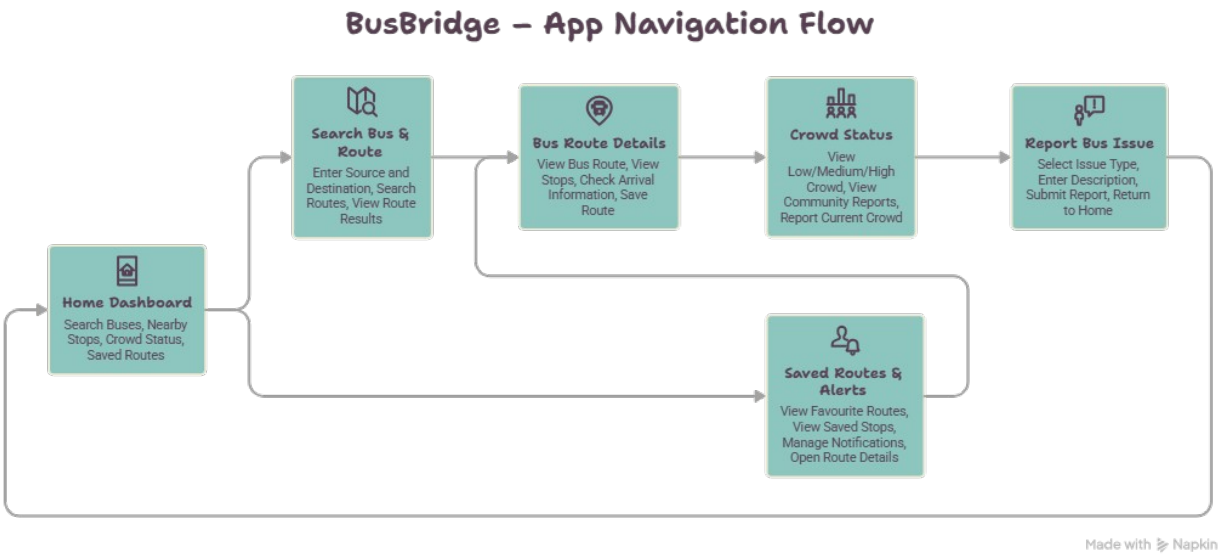


Figure 4: BusBridge navigation flow (prototype)

C. Glossary of Important Terms

Term	Meaning
MTC	Metropolitan Transport Corporation, the public bus operator in Chennai. BusBridge is not an official MTC application.

Term	Meaning
Concept / Prototype	A planned design and screen flow used to show how an app would work, without a complete working system.
Official Information	Information obtained from verified or authorized sources.
Estimated Information	Approximate information generated from available data.
Community-Reported	Information submitted by passengers, shown with its submission time.
GPS Feed	A stream of vehicle location data. Requires authorization and is not assumed to exist.
Crowd Status	A three-level indication (Low, Medium, High) of how crowded a bus is, as reported by passengers.
UI / UX	User Interface and User Experience: how an app looks and how easy it is to use.
Figma	A design tool used to create screens and interactive prototypes.
MVP	Minimum Viable Product: the smallest first version of an app that delivers its core value.
WCAG	Web Content Accessibility Guidelines published by W3C; used here as a reference for accessible design.
FCM	Firebase Cloud Messaging, a service for sending notifications.

D. Proposed Feature Summary

Feature	Description	Screen	Information Label
Bus Route Search	Find routes between two locations	2	Official / Estimated
Nearby Bus Stops	Discover stops near the passenger	1	Official
Route and Stop Details	View stops, boarding and destination	3	Official / Estimated
Arrival Information	Estimated arrival where data is supported	3	Official / Estimated / Unavailable
Live Location (conditional)	Shown only if authorized GPS data exists; otherwise unavailable	3	Official / Unavailable
Language Options	English and Tamil (proposed)	6	—
Crowd Status	View and submit Low / Medium / High reports	4	Community
Report Bus Issue	Report delays, cancellations, route changes	5	Community

Feature	Description	Screen	Information Label
Saved Routes	Save and manage favourite routes and stops	6	—
Service Alerts	Receive relevant alerts when supported	6	Official / Estimated